Here we replaced the third column with the average of the first two. It also makes the determinant of A equal to 0.

You could also try putting variable entries in the third column.

$$A = \begin{pmatrix} \frac{1}{6} & \frac{1}{9} & a \\ \frac{1}{3} & \frac{1}{3} & b \\ \frac{1}{2} & \frac{5}{9} & c \end{pmatrix}$$

Find the various choices for a, b, c which satisfy

$$|A| = 0$$

and

$$a + b + c = 1.$$

Taking the cross product $\mathbf{n} \times \mathbf{k}$ gives a vector which is along the line of intersection of two planes. It is clearly that any vector $\mathbf{w}$ which points in the steepest direction of plane P lies both in P and is perpendicular to the line of intersection of P and the xy-plane. Since $\mathbf{w}$ lies on P, it is perpendicular to $\mathbf{n}$.

and since it is perpendicular to the line of intersection of P and the xy-plane, it is perpendicular to $\mathbf{n} \times \mathbf{k}$. Thus

$$\mathbf{w} = (\mathbf{n} \times \mathbf{k}) \times \mathbf{n}$$

is a vector in P which points in the steepest direction of P.

(b) For the plane containing $\mathbf{u}$ and $\mathbf{v}$ we can take

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}$$

as a normal vector. Substituting this in the answer to part (a) we get

$$\mathbf{w} = \big((\mathbf{u} \times \mathbf{v}) \times \mathbf{k}\big) \times (\mathbf{u} \times \mathbf{v})$$

Notes:

(i) Obviously taking the products in different orders just introduces a scalar factor.

(ii) The formula for $\mathbf{w}$ does not give the zero vector (which would be bad), because the plane P is assumed not to be the xy-plane, so we are ensured that $\mathbf{n}$ and $\mathbf{k}$ form a positive angle.

Supplemental Problem Set 1
18.02 Exercises

Vectors and Matrices

1A: Vectors

Definition. A direction is just a unit vector. The direction of $\mathbf{A}$ is defined by

$$\operatorname{dir}\mathbf{A} = \frac{\mathbf{A}}{|\mathbf{A}|}, \qquad \mathbf{A} \neq 0.$$

1A–1 Find the magnitude and direction of the vector:

(a) $\mathbf{i} + \mathbf{j} + \mathbf{k}$

(b) $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

(c) $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

1A–2 For what values of c will

$$\frac{1}{5}\mathbf{i} - \frac{1}{5}\mathbf{j} + c\mathbf{k}$$

be a unit vector?

1A–3

(a) If $P = (1, 3, 1)$, find $\vec{OP}$, $|\,\vec{OP}\,|$, and $\operatorname{dir}\vec{OP}$.

(b) A vector $\mathbf{A}$ has magnitude 10 and direction

$$\frac{\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}}{3}.$$

If its tail is at $(-2, 0, 1)$, where is its head?

1A–4

(a) Let P and Q be two points in space, and X the midpoint of the line segment PQ. Let O be an arbitrary fixed point. Show that, as vectors,

$$\vec{OX} = \frac{1}{2}\left(\vec{OP} + \vec{OQ}\right).$$

(b) With the notation of part (a), assume that X divides the line segment PQ in the ratio $r : s$ where $r + s = 1$. Derive an expression for $\vec{OX}$ in terms of $\vec{OP}$ and $\vec{OQ}$.

1A–5 What are the $\hat{\imath}$ and $\hat{\jmath}$ components of a plane vector $\vec{A}$ of length 3, if it makes an angle of 30° with $\hat{\imath}$ and 60° with $\hat{\jmath}$? Is the second condition redundant?

1A–6 A small plane wishes to fly due north at 200 mph (as seen from the ground) in a wind blowing from the northeast at 50 mph. Tell with what vector velocity in the air it should travel (give the $\hat{\imath}$ and $\hat{\jmath}$ components).

1A–7 Let $\vec{A} = a\hat{\imath} + b\hat{\jmath}$ be a plane vector. Find, in terms of a and b, the vectors $\vec{A}'$ and $\vec{A}''$ resulting from rotating $\vec{A}$ by 90°:

(a) clockwise

(b) counterclockwise

(*Hint: Make the diagonal of a rectangle with sides on the x- and y-axes and rotate the whole rectangle.*)

1A–8 The direction (see definition above) of a space vector in engineering practice is often given by its *direction cosines.* To describe these, let

$$\vec{A} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$$

be a space vector represented as an origin vector, and let α, β, and γ be the three angles ($\leq \pi$) that $\vec{A}$ makes respectively with $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.

(a) Show that

$$\frac{\vec{A}}{|\vec{A}|} = \cos\alpha\,\hat{\imath} + \cos\beta\,\hat{\jmath} + \cos\gamma\,\hat{k}.$$

(The three coefficients are called the direction cosines of $\vec{A}$.)

(b) Express the direction cosines of $\vec{A}$ in terms of a, b, and c. Find the direction cosines of the vector $-\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.

(c) Prove that three numbers u, v, w are the direction cosines of a vector in space if and only if they satisfy

$$u^2 + v^2 + w^2 = 1.$$

1A–9 Prove using vector methods (without components) that the line segment joining the midpoints of two sides of a triangle is parallel to the third side and has half its length. (Call the two sides A and B.)

1A–10 Prove using vector methods (without components) that the midpoints of the sides of a space quadrilateral form a parallelogram.

1A–11 Prove using vector methods (without components) that the diagonals of a parallelogram bisect each other. (One way: let X and Y be the midpoints of the two diagonals; show that $X = Y$.)

1A–12 Label the four vertices of a parallelogram in counterclockwise order as O, P, Q, R. Prove that the line segment from O to the midpoint of PQ intersects the diagonal PR in a point X that is $\frac{1}{3}$ of the way from P to R.

(*Let* $\vec{A} = \vec{OP}$ *and* $\vec{B} = \vec{OR}$; express everything in terms of $\vec{A}$ and $\vec{B}$.)

1A-13

(a) Take a triangle PQR in the plane. Prove that the vectors

$$\vec{PQ} + \vec{QR} + \vec{RP} = 0.$$

(b) Continuing part (a), let $\vec{A}$ be a vector having the same length as $\vec{PQ}$ but perpendicular to it and pointing outside the triangle. Using similar vectors $\vec{B}$ and $\vec{C}$ for the other two sides, prove that

$$\vec{A} + \vec{B} + \vec{C} = 0.$$

(This only takes one sentence and no computation.)

1A-14

Generalize parts (a) and (b) of the previous exercise to a closed polygon in the plane which does not cross itself (i.e., one whose interior is a single region). Label its vertices $P_1, P_2, \ldots, P_n$ as you walk around it.

1A-15

Let $P_1, \ldots, P_n$ be the vertices of a regular n-gon in the plane, and O its center. Show without computation or coordinates that

$$\vec{OP_1} + \vec{OP_2} + \cdots + \vec{OP_n} = 0.$$

(a) If n is even.

(b) If n is odd.